BoredBrain AI

Project Utilities & Architecture Report

Generated: 2026-03-01 | Next.js 15.5 + React 19 | Vercel AI SDK 5.0 | TypeScript

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85+ UTILITY FILES	**66.8K** LINES OF CODE
28 AI TOOLS	**16** REACT HOOKS

Core Utilities 8 files

lib/utils.ts 148 lines

Core UI utilities — class merging, search group definitions, cache invalidation

- FN `cn(...inputs)` — Merge Tailwind classes (clsx + twMerge)
- FN `getSearchGroups(provider)` — Search group configs with icons
- FN `invalidateChatsCache()` — Dispatch cache invalidation event
- TYPE `SearchGroupId | SearchProvider`

`clsx` `tailwind-merge` `@hugeicons/core-free-icons`

lib/utils/with-retry.ts

54 lines

Exponential backoff retry + timeout-aware fetch

FN `withRetry<T>(fn, options)` — 2 retries, 500ms base, 30s timeout

FN `fetchWithTimeout(url, options)` — AbortController-based fetch (15s default)

lib/utils/tool-monitor.ts

85 lines

Tool execution monitoring — latency tracking, slow-tool warnings (>10s)

FN `monitorToolExecution<T>(name, fn)` — Wrap with timing + logging

FN `getToolStats()` — Aggregated per-tool stats (100-item buffer)

lib/errors.ts

205 lines

Centralized error handling — error codes, HTTP status mapping, action suggestions

CLASS `ChatSDKError` — Custom error with type, surface, status

FN `isAuthError | isRateLimited | isProRequired`

FN `getErrorActions(error)` — Suggested recovery actions

TYPE `ErrorType | ErrorCode | Surface`

lib/performance-cache.ts

118 lines

In-memory LRU cache with TTL — sessions (500/15min), usage counts (2000/5min)

CLASS `PerformanceCache<T>` — LRU cache with auto-cleanup (5min interval)

INST `sessionCache | usageCountCache`

FN `invalidateUserCaches(userId) | invalidateAllCaches()`

lib/logger.ts

53 lines

Environment-aware logging — sanitized output in production, stack trace redaction

CONST `logger` — .debug() .info() .warn() .error()

lib/parser.ts

113 lines

Markdown stream processor — buffers incomplete markdown patterns during streaming

FN `markdownJoinerTransform<TOOLS>()` — TransformStream for [text](url) and **bold** patterns

lib/constants.ts

15 lines

CONST `SEARCH_LIMITS` — DAILY: 5, EXTREME: 5

CONST `SNAPSHOT_NAME` — Daytona sandbox snapshot ID

🔒 Authentication & User Data 5 files

lib/auth.ts

100+ lines

Better Auth config — OAuth (GitHub, Google, Twitter), Drizzle adapter, rate limiting (50/60s)

lib/auth-utils.ts

42 lines

FN `getSession(input?)` — Cached session retrieval

FN `getUser()` — Get authenticated user

lib/auth-client.ts

11 lines

CONST `authClient | useSession | signIn | signOut | signUp`

lib/user-data-server.ts

100+ lines

Server-side user data — subscription, payments, referrals, pro status, guest handling

TYPE `ComprehensiveUserData`

FN `ensureGuestUserRecord() | buildGuestComprehensiveData()`

lib/agent-api/auth.ts

145 lines

API Key auth — bb_sk_ prefix, Redis rate limiting (60s window), in-memory fallback

FN `generateApiKey()` — crypto.randomBytes(32)

FN `validateApiKey(key)` — DB lookup + last used update

FN `checkRateLimit(keyId, limit)` — Redis INCR+EXPIRE

FN `authenticateRequest(req)` — Full auth flow

Database & ORM

3 files

lib/db/index.ts

36 lines

Drizzle ORM — Neon HTTP driver, Upstash Redis cache (600s TTL), 2 read replicas

CONST `maindb | db` — Main + withReplicas

lib/db/schema.ts

370+ lines

17 tables — user, session, chat, message, agent, promptTemplate, arenaMatch, etc.

lib/db/queries.ts

100+ lines

DB queries — user CRUD, chat pagination, message management, usage tracking, $withCache()

AI Providers & Models 1 file

ai/providers.ts

262 lines

Multi-provider model registry with access control, vision/reasoning flags, pricing tiers

MODEL ID	PROVIDER	TIER	VISION	REASONING
boredbrain-default	xAI Grok 4 Fast	Free	✓	
boredbrain-grok-3	xAI Grok 3	Pro		
boredbrain-grok-4	xAI Grok 4	Pro	✓	✓
boredbrain-grok-4-fast-think	xAI Grok 4 Fast	Pro	✓	✓
boredbrain-code	xAI Grok 4 Fast	Pro		✓
boredbrain-google-lite	Gemini 2.5 Flash Lite	Free	✓	
boredbrain-google	Gemini 2.5 Flash	Pro	✓	
boredbrain-google-pro	Gemini 2.5 Pro	Pro	✓	

AI Tool Implementations 28 tools

TOOL NAME	CATEGORY	PRICE (BBAI)	RATE LIMIT	FILE
web_search	search	10	30/min	lib/tools/web-search.ts
extreme_search	search	50	5/min	lib/tools/extreme-search.ts
x_search	search	20	20/min	lib/tools/x-search.ts
academic_search	search	15	20/min	lib/tools/academic-search.ts
reddit_search	search	10	20/min	lib/tools/reddit-search.ts
youtube_search	search	10	20/min	lib/tools/youtube-search.ts

TOOL NAME	CATEGORY	PRICE (BBAI)	RATE LIMIT	FILE
stock_chart	finance	10	30/min	lib/tools/stock-chart.ts
coin_data	finance	5	60/min	lib/tools/crypto-tools.ts
coin_ohlc	finance	5	60/min	lib/tools/crypto-tools.ts
wallet_analyzer	finance	30	10/min	lib/tools/wallet-analyzer.ts
nft_retrieval	finance	8	30/min	lib/tools/nft-retrieval.ts
token_retrieval	finance	8	30/min	lib/tools/token-retrieval.ts
currency_converter	finance	3	60/min	lib/tools/currency-converter.ts
code_interpreter	utility	25	10/min	lib/tools/code-interpreter.ts
text_translate	utility	5	60/min	lib/tools/text-translate.ts
retrieve	utility	8	30/min	lib/tools/retrieve.ts
datetime	utility	1	120/min	lib/tools/datetime.ts
find_place_on_map	location	5	30/min	lib/tools/map-tools.ts
nearby_places_search	location	5	30/min	lib/tools/map-tools.ts
get_weather_data	location	3	60/min	lib/tools/weather.ts
track_flight	location	8	20/min	lib/tools/flight-tracker.ts
movie_or_tv_search	media	5	30/min	lib/tools/movie-tv-search.ts
trending_movies	media	3	30/min	lib/tools/trending-movies.ts
trending_tv	media	3	30/min	lib/tools/trending-tv.ts

React Hooks 16 hooks

hooks/use-signals.ts 408 lines

Full signals/lookouts management — CRUD, optimistic updates, polling (5s), completion detection

HOOK `useSignals()` — Create, update, delete, test signals

HOOK `useFilteredSignals(filter)` — Filter by status

hooks/use-cached-user-data.tsx

141 lines

HOOK `useCachedUserData()` — Full user data + localStorage cache

HOOK `useCachedIsProUser()`

HOOK `useCachedSubscriptionStatus()`

hooks/use-user-data.ts

121 lines

HOOK `useUserData()` — React Query (30min stale)

HOOK `useIsProUser() | useSubscriptionStatus()`

hooks/use-local-storage.tsx

88 lines

HOOK `useLocalStorage<T>(key, default)` — SSR-safe, cross-tab sync

hooks/use-usage-data.ts

15 lines

HOOK `useUsageData(user, enabled?)` — Message count (1min stale)

hooks/use-optimized-scroll.ts

32 lines

HOOK `useOptimizedScroll(ref)` — scrollToBottom, markManualScroll, resetManualScroll

Other Hooks

10 more

HOOK `useAutoResume` — Resume interrupted streams

HOOK `useChatVisibility` — Public/private toggle

HOOK `useChatBanners` — Auto-dismiss banners (3.5s)

HOOK `useSuggestedQuestions` — Follow-up suggestions

HOOK `useSettingsDialog` — Settings + URL hash sync

HOOK `useModelAutoSwitch` — Auto-downgrade pro model

HOOK `useIsMobile` — Viewport <768px

HOOK `useLocation` — Geolocation detection

HOOK `useMediaQuery` — Reactive matchMedia

HOOK `useGitHubStars` — Repo star count

External API & ENV Status .env.local

Connected (Infrastructure)

DATABASE_URL	Neon PostgreSQL
REDIS_URL	Redis
UPSTASH_REDIS_*	Upstash Cache
BETTER_AUTH_SECRET	Auth
GITHUB_CLIENT_*	GitHub OAuth
GOOGLE_CLIENT_*	Google OAuth
TWITTER_CLIENT_*	Twitter OAuth
BLOB_READ_WRITE_TOKEN	Vercel Blob
QSTASH_TOKEN	QStash Cron
RESEND_API_KEY	Email

Dummy (Not Connected)

XAI_API_KEY	Grok 3/4
GOOGLE_GENERATIVE_AI_*	Gemini
OPENAI_API_KEY	GPT (unused)
ANTHROPIC_API_KEY	Claude (unused)
GROQ_API_KEY	Groq (unused)
DAYTONA_API_KEY	Code Interpreter
TAVILY_API_KEY	Web Search
EXA_API_KEY	Web Search
FIRECRAWL_API_KEY	Web Crawling
COINGECKO_API_KEY	Crypto Data
TMDB_API_KEY	Movie/TV
OPENWEATHER_API_KEY	Weather
ELEVENLABS_API_KEY	Audio TTS
SMITHERY_API_KEY	MCP Server

💰 Agent Economy (Smart Contracts) 3 contracts

contracts/contracts/BBAIToken.sol ERC20

$BBAI Token — 1B supply, vesting, distribution (40% ecosystem, 20% team, 15% liquidity, 15% community, 10% investors)

STATUS Defined only — NOT DEPLOYED, NOT INTEGRATED with backend

contracts/contracts/AgentRegistry.sol

ERC721

Agent NFT Registry — 100 BBAI registration, on-chain agent ownership

STATUS Defined only — NOT DEPLOYED

contracts/contracts/PaymentRouter.sol

Router

Payment Router — 85% agent owner / 15% platform fee split, batch payments

STATUS Defined only — NOT DEPLOYED

Middleware & Routing

1 file

middleware.ts

87 lines

Edge middleware — session cookie check, route protection, guest access flag

PUBLIC /api/tools, /api/agents, /api/arena, /api/keys, /.well-known, /arena, /agents, /stats

PROTECTED /api/search, /api/prompts — require session auth

FLAG ALLOW_GUEST_ACCESS — bypass auth when true

BoredBrain AI — AI Agent Economy Platform

85+ utility files • 66,800+ lines • 28 AI tools • 17 DB tables

https://boredbrain-master.vercel.app